

CE725: Vibration-Based Structural Health Monitoring

Take Home End Semester Examination – Part I

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Introduction

This report presents the solutions of Question 1 and Question 2 of the assignment. The tasks involved system identification using Markov parameters and ERA, modal parameter extraction, and structural health monitoring using identified state-space models.

Question 1: 5-DOF System Identification

A five degree-of-freedom lumped mass system was excited using a known input force and acceleration responses at all five DOFs were measured at a sampling rate of 10 Hz.

(a) Identification of Markov Parameters

The first 100 Markov parameters were estimated using the direct least-squares approach.

A Toeplitz input matrix U was formed from the force history and the Markov parameter matrix was obtained using:

$$M = U^\dagger Y$$

where $U^\dagger$ is the Moore-Penrose pseudoinverse.

The identified Markov matrix had dimensions:

$$5 \times 100$$

(b) Comparison with True Markov Parameters

The identified Markov parameters were compared with the true values provided in the file `Q1_TrueMarkovPara.mat`. Good agreement was observed.

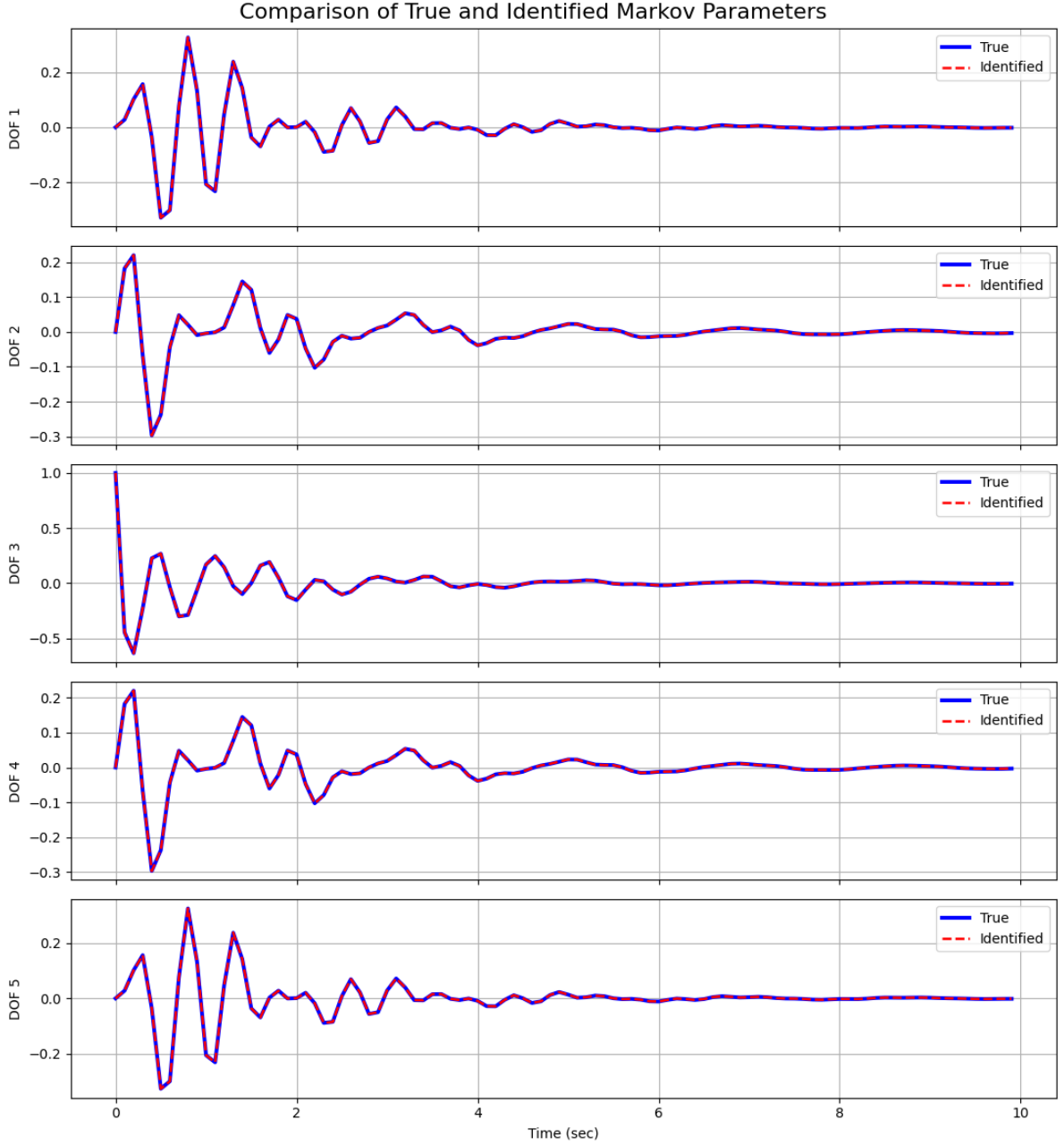


Figure 1: Comparison of Markov Parameters

(c) ERA State Space Model

The Eigensystem Realization Algorithm (ERA) was used to obtain a discrete-time state-space model.

Hankel matrices were formed using:

$$r = 40, \quad s = 40$$

Singular Value Decomposition of H_1 :

$$H_1 = U\Sigma V^T$$

A sharp drop in singular values was observed after the 6th singular value, hence:

$$n = 6$$

was selected as model order.

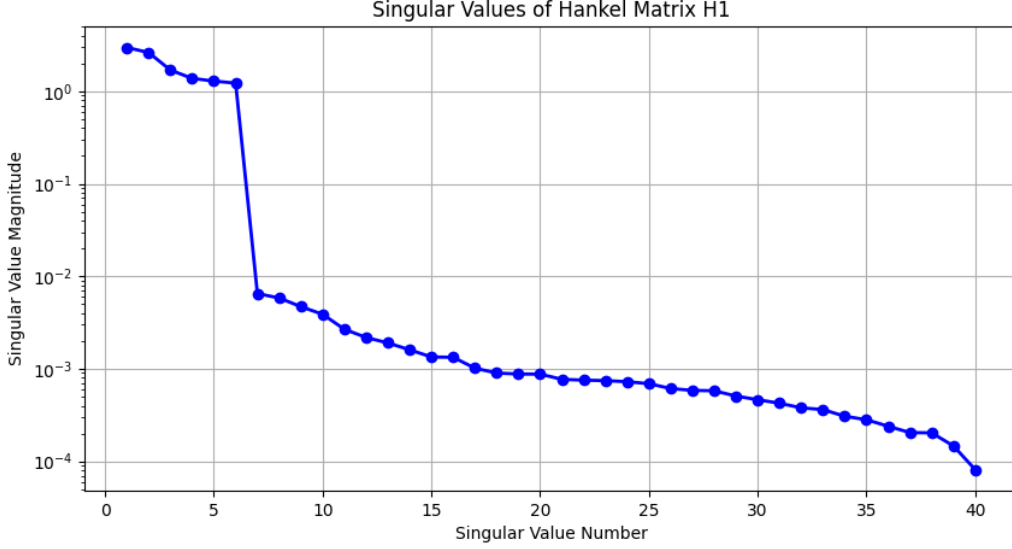


Figure 2: Singular Values of H_1

(d) Complex Modal Parameters

From the identified state matrix A_d , eigenvalues and eigenvectors were computed.
Discrete poles:

$$\lambda_i = eig(A_d)$$

Continuous poles:

$$s_i = \frac{\ln(\lambda_i)}{\Delta t}$$

Complex mode shapes were obtained using:

$$\psi_i = C_d \phi_i$$

(e) Real Modal Parameters

Real modal parameters were extracted from the complex modes.
The identified frequencies and damping ratios were:

Mode	Frequency (Hz)	Damping Ratio
1	0.543	0.0998
2	1.686	0.0800
3	2.229	0.0500

Table 1: Identified Modal Parameters

(f) Comparison with True Modal Parameters

The identified modes corresponded to Modes 1, 3 and 5 of the true system.

All Modal Assurance Criterion (MAC) values were close to 1, indicating excellent mode shape agreement.

Mode	Matched True Mode	% Error f	% Error ζ	MAC
1	1	< 1%	< 1%	1.00
2	3	< 1%	< 1%	1.00
3	5	< 1%	< 1%	1.00

Table 2: Comparison with True Modal Parameters

(g) Bonus Question

Only one external force was applied. If force is applied at one DOF, modes having small modal amplitude at that DOF receive little energy. So those modes barely appear in output data. This means their Markov contribution is tiny.

Question 2: Structural Health Monitoring of a Two-Storey Building

State-space matrices identified from forced vibration tests on three successive days were provided. Day 1 was treated as the healthy reference condition.

Sampling frequency:

$$f_s = 100 \text{ Hz}$$

Methodology

For each day:

1. Eigenvalues of A_d were computed.
2. Continuous poles were obtained.
3. Natural frequencies and damping ratios were extracted.
4. Mode shapes were obtained from C_d .
5. Day 2 and Day 3 were compared with Day 1 using:
 - Frequency shift
 - Damping change
 - MAC values

Results

Mode	Day1 (Hz)	Day2 (Hz)	Day3 (Hz)	MAC Day2	MAC Day3
1	0.311	0.311	0.272	1.00	0.974
2	0.814	0.814	0.658	1.00	0.974

Table 3: Comparison of Modal Frequencies

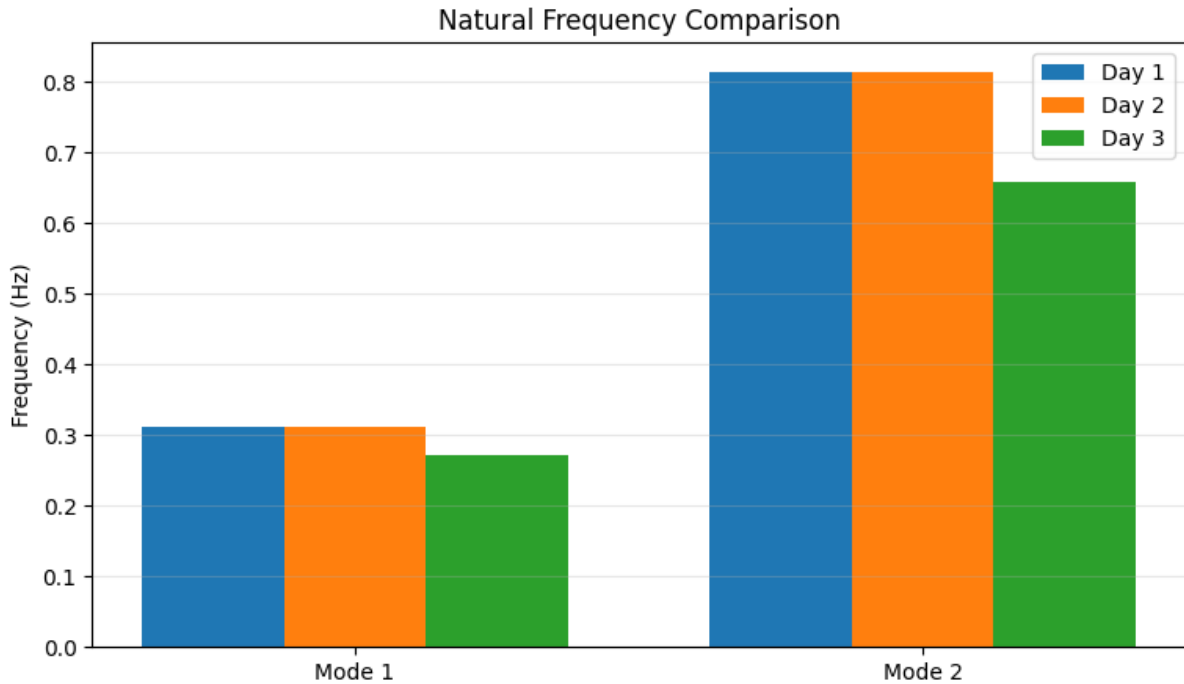


Figure 3: Natural Frequency Comparison for Three days

Interpretation

Day 2: Frequencies remained unchanged and MAC values were equal to 1.0. Therefore, the building remained healthy.

Day 3: Frequencies reduced significantly by approximately 12% and 19%, with slight reduction in MAC values. This indicates reduction in structural stiffness and hence structural damage.

Conclusion

The ERA method successfully identified the dynamic characteristics of the 5-DOF system. Accurate modal parameters were obtained with very small errors.

For the two-storey building, Day 2 remained healthy, while Day 3 showed clear signs of structural degradation based on frequency reduction and mode shape changes.